

When Is Reuse Safe? Regime-Aware Policy Transfer for Recurring Drift in Network Telemetry

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Abstract—Network telemetry drifts, and the standard fix, retraining, pays the full cost of rebuilding the model every time. But network regimes recur: a congestion profile, a radio configuration or an operator comes back. We ask whether a policy learned for a regime can be reused when that regime returns, and when doing so is safe. Regime-Aware Policy Transfer (RAPT) stores one ensemble policy per regime, retrieves it on recurrence, and refits it when rolling accuracy shows it no longer fits. On four network streams under a strict prequential protocol, base RAPT uses 48–92% less adaptation CPU than full retraining, though the cost-aware variant that recovers accuracy saves less. Macro-F1 is lower on all four streams, significantly so on two. UGR’16 is the clearest failure: base RAPT reaches 0.836, and refitting a reused policy on a larger recent buffer recovers it to 0.928. The parity check intended to trigger that refit never fires at any threshold we tried, so the recovery is the buffer’s, not the check’s. Our tests do not settle whether the UGR’16 failure is a changed feature-to-label mapping or a covariate shift. On 5G Campus, a cost-aware refit configuration matches full retraining within a ± 0.010 macro-F1 margin ($\Delta = +0.0022$, $p = 0.31$) at about 40% less adaptation CPU, though a periodic cheap refit matches it on both accuracy and cost. Two other mechanisms failed under measurement and we report them: a relative-degradation trigger that cannot fire on a policy that was poor from the start, and an incremental learner that was worse on both accuracy and cost.

Index Terms—concept drift, network telemetry, online learning, adaptive ensembles, regime-aware learning, computationally efficient adaptation

I. INTRODUCTION

The data seen by network monitoring systems does not stay the same for long. Load, radio conditions, subscriber mobility and configuration changes each move the distribution, and a classifier fitted on a period of a 5G testbed or an ISP netflow archive will not stay accurate for very long. Retraining once performance drops is the standard answer. It works, but it has to be paid for again every time, and with high-rate telemetry the total spent on rebuilding models may end up above what it costs to serve predictions. Lightweight detection and adaptation frameworks for constrained devices already exist [27], and many good detectors decide *when* to adapt [2], [4], [5], [8], [10], [19], [22], but they make no use of how the change is structured. Conditions in networks often come back. A weekday congestion profile, a radio configuration or an operator-specific QoS regime may go away and return later. Some of the earliest work in the field was driven by recurring concepts [23], [25], and still the usual practice is to learn again

from scratch, even though the model being thrown away may well be the right one.

The question we ask here is whether the discarded information can be kept and used. RAPT keeps a repository of adaptation policies, each indexed by an operating regime. At a regime boundary it looks up whether the incoming regime already has a stored policy. When it does, that ensemble is retrieved at close to zero adaptation cost, and when it does not, a new policy is trained and added to the repository. Since a retrieved policy can be wrong if the label relationship in that regime has changed, RAPT also follows its streaming accuracy and refreshes the policy if needed.

If regimes recur with stable label semantics, reuse keeps accuracy and removes most of the adaptation cost. If a reused policy no longer fits the returning regime, the accuracy that reuse was meant to protect suffers; in the one stream where that happens (UGR’16) the repair that works is refitting on a larger recent buffer, not the parity check. Our contributions are as follows:

- a regime-conditioned policy-transfer mechanism that stores and retrieves ensemble policies by regime;
- a monitoring and refresh strategy whose cost-aware configuration matches full-retraining accuracy at a fraction of the cost;
- an evaluation on four network streams under one prequential protocol, against four established drift detectors;
- a failure analysis covering harmful blind reuse, a trigger that cannot fire by construction, and an incremental learner that loses on both axes.

II. RELATED WORK

A. Concept and Data Drift

Drift in a data stream is usually sorted into two cases. Under covariate shift the marginal $P(X)$ changes while $P(Y | X)$ holds, and under concept drift it is the conditional $P(Y | X)$ that changes [10], [15]. Surveys have catalogued the resulting taxonomies [13]. The story starts with STAGGER and its recurring concepts [23], and with Widmer’s analysis of robustness against flexibility [25].

B. Drift Detection

ADWIN compares sub-window means inside an adaptive window [4], and Page-Hinkley is a sequential cumulative-sum procedure [19]. DDM and EDD follow the error rate

together with its variance [2], [10]. RDDM was designed to give fewer false alarms [3], while the EDMA and EWMA charts track an exponentially weighted moving average of the error [22]. A separate family needs no labels and tests the features directly, using Hoeffding bounds [9], an incremental Kolmogorov–Smirnov statistic [8], or the earth-mover distance [5], with lightweight versions for constrained IoT deployments [27].

C. Adaptive and Online Learning

Online ensembles handle drift by updating their members as new data arrives. Dynamic weighted majority weights each learner by its recent accuracy and adds or drops learners along the way [14]; it grew out of earlier methods that minimize disagreement on drifting concepts [12]. Adaptive random forests pair online bagging with drift-aware tree replacement [11], and the accuracy-updated ensemble reweights its components when drift appears [6]. Similar designs have been proposed for social big-data and IoT streams [1], [26]. Single-pass incremental learners, of which Hoeffding trees are the standard example, keep memory use and per-example update cost bounded [7]. The detectors and incremental learners in our experiments come from the River library [18]. All of this work is about *how* a model gets updated; none of it considers fetching a model that is already good enough. In Section V-G we test one incremental learner against our approach, and on our stream it does not improve the accuracy–cost trade-off.

D. Transfer under Recurring Conditions

Transfer learning moves representations or parameters learned on a source task over to a related target task [21], with roots in work on the role of forgetting [16] and on the cost of reusing compiled knowledge chunks [24]. RAPT saves a fitted policy with its regime identity and retrieves the two together. The decision it makes is whether any adaptation is needed; how to adapt is left aside. With no recurrence, the mechanism collapses into ordinary retraining.

III. PROBLEM FORMULATION

We treat the data as a chronological stream $x_1, x_2, \dots$, where $x_t \in \mathbb{R}^d$ is a feature vector and $y_t \in \{1, \dots, C\}$ is its label. The label is unknown until the prediction for x_t has been issued. The stream is divided into fixed-size windows $W_1, \dots, W_T$, and each window W_k is tagged with a regime identifier $r_k \in \mathcal{R}$. A regime is a semantically defined operating condition, such as a network scenario or a time-of-day block. It is not a cluster label. A *policy* π is a fitted model plus its decision rule, and the repository is a mapping $\mathcal{P} : \mathcal{R} \rightarrow \pi$ that is filled in as new regimes show up. Whenever the stream moves from r_{k-1} to r_k , the controller picks an action $a_k \in \{\text{REUSE}, \text{TRAIN}\}$. It reuses $\mathcal{P}(r_k)$ when that exists, and trains a new policy on the recent buffer when it does not. Let $M(\pi, W)$ be a performance measure, which is macro-F1 in this paper, and let $c(\pi)$ be the cost of producing π in adaptation CPU time. The objective is to hold $M(\cdot, W_k)$ near

the level a freshly trained policy would reach while making $\sum_k c(a_k)$ as small as possible.

Reuse rests on an assumption, namely that $P(Y | X, r_k)$ has not changed since the policy was stored. If the regime identifier returns after the conditional relationship has shifted, the stale policy gets applied and nothing flags it. The hard part is separating a recurring regime from a recurring label relationship.

IV. PROPOSED METHOD: RAPT

A. Architecture and Repository

RAPT sits between the stream and the model and makes one decision at each regime boundary. The active policy predicts each window, and the window is scored when its labels arrive. A change of policy happens only at a regime transition or when monitoring reports degradation.

We represent a regime with its semantic identifier and do not learn a clustering. What we give up is that the mechanism requires a usable regime signal; see Section VIII. The repository holds one checkpoint per regime, made up of the fitted ensemble, its soft-voting weights, and timestamps for creation and last access. Checkpoints are immutable.

B. Matching and Transfer

Matching uses the regime identifier. On a transition to regime r , the controller tests whether $r \in \mathcal{P}$. The enhanced variant also applies a fingerprint-similarity gate, which requires the feature summary of the current window to be close to the fingerprint saved when the checkpoint was created. The effect is to relax the question “has this regime been seen?” into “has a sufficiently similar instance been seen?”

When reuse goes ahead, the retrieved ensemble becomes the active policy, and its soft-voting weights can be recalibrated using the most recent buffered window. Calibration computes the recent accuracy of the two components and renormalizes their weights. It covers at most the last 50 observations and is skipped when fewer than 20 are available, so its cost is small and fixed. If no checkpoint exists, we train a new policy from a class-anchored buffer and store it. The buffer holds a stratified slice of the training prefix. All adaptive models here use it in the same way, which stops a refit from collapsing onto a single class when the recent window has only one.

C. Monitoring, Refresh, and Fallback

We use two monitoring rules, and how they differ is one of our findings. The *absolute* rule triggers when rolling accuracy goes under a fixed floor. The *relative* rule triggers when accuracy has dropped by more than a fixed margin from a decayed reference, which is built from that policy’s own earlier accuracy. A relative rule cannot catch a policy that was never good, because the reference decays toward the poor level and so the gap it looks for never appears.

A refresh rebuilds the active policy from the recent buffer and replaces its checkpoint, and any later reuse then gets the refreshed policy. In the default refresh, the 100-tree ensemble is refit on a 1000-observation buffer. The cost-aware refresh

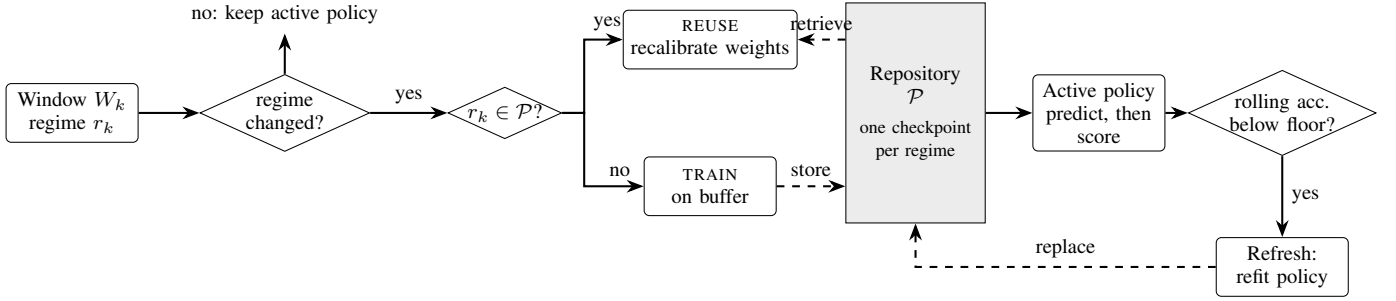


Fig. 1. RAPT overview. At each regime boundary the controller looks up the incoming regime in the repository and either reuses the stored policy or trains and stores a new one. The active policy predicts and is scored as labels arrive. A monitor refits the policy and replaces its checkpoint when rolling accuracy degrades. Solid arrows are control flow, dashed arrows are repository reads and writes.

has the same trigger but refits 20 trees on a 300-observation buffer. RAPT-Enhanced contains two additions. *Buffer-blended novelty refitting* refits a novel regime on a larger recent buffer, requested at 1500 observations against 500 for the original; the adaptation buffer is capped at 1000, so on UGR’16 the refit effectively uses about 1000 rows (Section V). *Selective parity refitting* was intended to refit a reused policy and replace its checkpoint whenever rolling accuracy falls under a parity threshold over a short window. At the configured threshold this branch never activates in any of our runs (parity_refits = 0 on every stream and seed), so it contributes nothing to the reported numbers. When there is no checkpoint, or the similarity gate turns down the one that exists, the controller trains a fresh policy and stores it under the incoming regime identifier.

V. EXPERIMENTAL METHODOLOGY

A. Datasets

Four datasets are summarized in Tab. I; all are real network traces. *5G Campus* is a testbed QoS stream with delay metrics labelled GOOD, DEGRADED or BAD from the 90th-percentile delay of the next window; each of the three scenarios recurs fourteen times. *UGR’16* is aggregated ISP netflow with 134 features. Because the calibration split contains almost no labelled attack traffic, we use the first thirty days of labelled data, one sample per minute (43,200 samples); time-of-day crossed with weekday/weekend gives twelve regimes and 168 recurrences. *NordicDat* is a cross-border LTE/5G predictive-QoS trace [17] whose regimes are operators. Both of these label the median delay of the next window into three classes. *5G NR* is a simulated end-to-end latency stream [20] of 499 windows of 500 packets across four regimes (ABCDACBDAB), labelled with the same rule as Campus.

B. Preprocessing, Ordering, and Leakage Control

All four streams undergo the exact same processing. First, all four streams undergo sorting by row according to timestamp. Then, all identifier and constant columns are dropped. Finally, only those features that are numeric are retained. Any infinite values present within a stream are treated as missing. Any rows that lack a target value are deleted. Any

TABLE I
DATASETS USED IN EVALUATIONS. RECURRENCES DENOTE OCCURRENCES WHERE A PARTICULAR REGIME IS REVISITED.

Property	Campus	UGR’16	Nordic	5G NR
Samples	1,799	43,200	91,455	249,500
Features	19	134	16	12
Classes	3	2	3	3
Windows	179	180	182	499
Window size	10	240	500	500
Regimes	3	12	3	4
Recurrences	14	168	17	6

remaining gaps are then filled using medians computed solely from the training prefix. Target thresholds and any scale-related statistics also derive from that prefix. None of the streams are shuffled. The prefix is the first twenty percent of windows: thirty-six for the three cross-dataset streams and ninety-nine for 5G NR. That prefix is completely excluded from evaluation, and that is the sole source of target thresholds. The order of operations is predict → evaluate → adapt for every window prior to its labels being used for adaptation. Regime annotation occurs directly from reading the stream. Therefore, no derivation of annotation can occur based upon the target.

C. Models And Detectors

Each of the five primary models utilizes exactly the same base architecture and the same retraining buffer. Therefore, any variation between them is due to their respective adaptation mechanisms only. The base is an ensemble of a random forest classifier and an extra-trees classifier, both having fifty trees of depth seven. Frozen is trained once on the prefix and receives no subsequent updates. Event-Driven will update when window accuracy falls below a fixed threshold. Full Retraining will recreate/rebuild the model at every regime transition. RAPT and RAPT-Enhanced represent the controller(s) explained in Section IV.

ADWIN, Page-Hinkley, EDD and EDMA operate within the same retraining harness, with identical model, buffer, capacity and minimum-retrain-interval. They differ only by how they detect triggers. We pre-set their parameters in advance and did

not adjust them via testing against our test streams (ADWIN $\delta = .002$; Page-Hinkley min = Thirty; THRESHOLD = Fifty; EDD WARNING = Two; DRIFT = Three; EDMA $\alpha = .2$ k = Two). Since none of our streams include true ground truth drift boundary indicators we report detected events, adaptation costs and predictive performance. Precision and Recall were not reported for our detectors.

D. Metrics And Statistics

Performance is evaluated through macro-F1, Accuracy, Precision and Recall measures. Costs may be measured in two fashions. Adaptation CPU denotes CPU time exclusively devoted to adaptation routines while Runtime indicates Total Streaming Time. A reduction in Adaptation Time does not necessarily translate into a proportional reduction in Total Runtime.

Each experiment had seeds [42,43,44,45,46]. Therefore we report Mean and Standard Deviation for each measure. For comparison purposes we use the Wilcoxon Signed-Ranked Test for paired comparisons along with Cohen's d. With respect to pairing by Window there would be approximately 143–146 observations. When paired by Seed we would have $n = 5$ with a minimum two-sided p-value available to us of .0625. If a result shows no significance we state that no difference was demonstrated between methods.

E. Predictive Performance

Table II holds the primary comparison. On UGR'16, RAPT tracks the frozen baseline and not the retrained models, the signature of stale policy reuse; on 5G NR the five models are within 0.0133 of one another.

On 5G Campus every model is close to saturation. Full Retraining has the best macro-F1 (0.9836 ± 0.0013), while RAPT and RAPT-Enhanced both land at 0.9381 ± 0.0016 . That is 0.0455 lower, and the gap is significant ($p = 2.6 \times 10^{-4}$, $d = -0.32$).

UGR'16 is where the assumptions of the mechanism fail, and it teaches us the most. Frozen has the highest macro-F1 (0.9690 ± 0.0029), which is consistent with attacks being sparse and clustered. RAPT reaches only 0.8360 ± 0.0262 : 0.1236 below Full Retraining ($p = 2.6 \times 10^{-11}$, $d = -0.67$) and the worst of the five models, despite its 133 reuses. The twelve regimes do recur and their identifiers stay valid through the 30-day block. Two readings of the deficit are consistent with our data and we do not separate them: the refit buffer is too small, and the buffer a policy is stored under can belong to the previous regime. RAPT-Enhanced, which refits on a larger recent buffer, recovers macro-F1 to 0.9276 ± 0.0123 ; that gain of 0.0917 ($p = 5.4 \times 10^{-9}$, $d = 0.56$) is the one statistically supported improvement of a variant over another on this stream, and its parity branch never activates, so the recovery is the buffer's. Re-scoring pooled over all evaluation samples shifts the ranking (Frozen 0.8216, RAPT 0.6629) and RAPT's false-alarm rate on attack-free windows is two orders of magnitude above Frozen's, so we treat UGR'16 per-window macro-F1 as metric-sensitive.

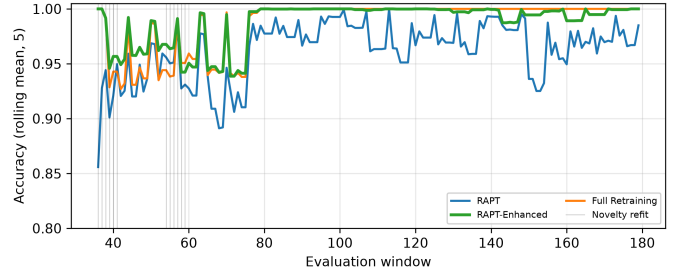


Fig. 2. Rolling accuracy over evaluation windows on UGR'16. RAPT sags once reused policies go stale, and refitting on a larger recent buffer (RAPT-Enhanced) pulls accuracy back toward Full Retraining. Vertical ticks mark novelty refits: the first window of each new regime, where RAPT trains a fresh policy.

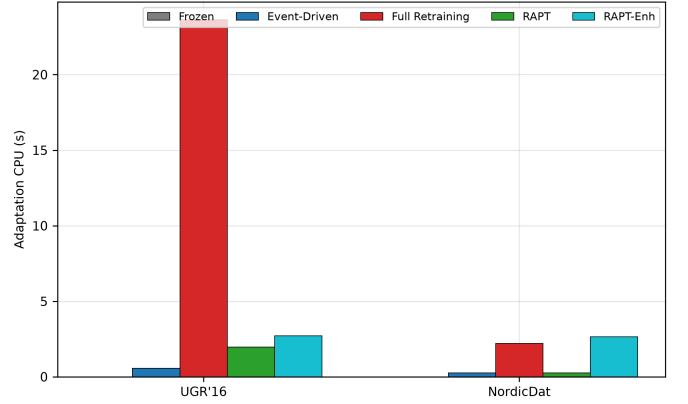


Fig. 3. Adaptation CPU on the two largest streams. The ordering is the reverse of macro-F1: on UGR'16, RAPT uses an order of magnitude less adaptation time than Full Retraining and records lower macro-F1.

NordicDat is hard to predict. Frozen scores 0.2779 ± 0.0224 , adaptation helps, and Full Retraining gets 0.4227 ± 0.0299 against 0.3818 ± 0.0184 for RAPT. The deficit of 0.0409 is not significant ($p = 0.255$, $d = -0.12$). RAPT does have the highest accuracy on this stream (0.6172 ± 0.0127 against 0.5694 ± 0.0075). On 5G NR the five models are only 0.0133 apart. Taken together, these results give no basis for saying RAPT improves accuracy: Full Retraining has the best macro-F1 on three of four streams, and RAPT sits below it on all four. The case for RAPT rests on cost and on knowing when reuse is safe.

F. Computational Cost and Policy Reuse

Table II also lists adaptation and total cost, and Fig. 3 compares adaptation CPU. Relative to Full Retraining the cost falls on every stream: $1.4008 \rightarrow 0.2292$ s on 5G Campus (-83.6%), $23.6318 \rightarrow 1.9711$ s on UGR'16 (-91.7%), $2.2034 \rightarrow 0.2594$ s on NordicDat (-88.2%), and $0.7636 \rightarrow 0.3955$ s on 5G NR (-48.2%). Full Retraining retrains at every regime transition (14, 144, 18 and 9 times); RAPT retrains 2, 11, 2 and 3 times and retrieves a stored policy 12, 133, 16 and 6 times.

TABLE II

PRIMARY COMPARISON AND COST, MEAN OVER SEEDS 42–46. BOLD MARKS THE BEST MACRO-F1 PER STREAM; CPU AND RUNTIME IN SECONDS, EVENTS ARE COUNTS.

Dataset	Model	F1	Acc.	Prec.	Rec.	CPU (s)	Run (s)	Retr.	Reuse
5G Campus	Frozen	0.9364 $\pm$ 0.0023	0.9822 $\pm$ 0.0006	0.9379 $\pm$ 0.0025	0.9377 $\pm$ 0.0017	0.0000	1.2482	0	–
	Event-Driven	0.9636 $\pm$ 0.0006	0.9908 $\pm$ 0.0003	0.9637 $\pm$ 0.0008	0.9646 $\pm$ 0.0002	0.1080	1.3926	1	–
	Full Retraining	0.9836 $\pm$ 0.0013	0.9951 $\pm$ 0.0000	0.9838 $\pm$ 0.0015	0.9843 $\pm$ 0.0012	1.4008	2.6677	14	–
	RAPT	0.9381 $\pm$ 0.0016	0.9838 $\pm$ 0.0003	0.9393 $\pm$ 0.0017	0.9392 $\pm$ 0.0016	0.2292	1.4931	2	12
	RAPT-Enhanced	0.9381 $\pm$ 0.0016	0.9838 $\pm$ 0.0003	0.9393 $\pm$ 0.0017	0.9392 $\pm$ 0.0016	0.2337	2.1046	2	12
UGR'16	Frozen	0.9690 $\pm$ 0.0029	0.9899 $\pm$ 0.0007	0.9795 $\pm$ 0.0030	0.9662 $\pm$ 0.0028	0.0000	2.4914	0	–
	Event-Driven	0.8972 $\pm$ 0.0277	0.9898 $\pm$ 0.0013	0.9042 $\pm$ 0.0269	0.8955 $\pm$ 0.0280	0.5754	3.1303	3	–
	Full Retraining	0.9595 $\pm$ 0.0012	0.9873 $\pm$ 0.0007	0.9726 $\pm$ 0.0014	0.9580 $\pm$ 0.0015	23.6318	26.2627	144	–
	RAPT	0.8360 $\pm$ 0.0262	0.9635 $\pm$ 0.0182	0.8527 $\pm$ 0.0205	0.8336 $\pm$ 0.0294	1.9711	4.4990	11	133
	RAPT-Enhanced	0.9276 $\pm$ 0.0123	0.9880 $\pm$ 0.0017	0.9375 $\pm$ 0.0117	0.9251 $\pm$ 0.0123	2.7182	5.9329	11	133
NordicDat	Frozen	0.2779 $\pm$ 0.0224	0.4770 $\pm$ 0.0123	0.4215 $\pm$ 0.0226	0.2487 $\pm$ 0.0229	0.0000	2.5488	0	–
	Event-Driven	0.2547 $\pm$ 0.0177	0.4583 $\pm$ 0.0367	0.3909 $\pm$ 0.0208	0.2274 $\pm$ 0.0151	0.2533	2.8435	2.2	–
	Full Retraining	0.4227 $\pm$ 0.0299	0.5694 $\pm$ 0.0075	0.4938 $\pm$ 0.0233	0.4085 $\pm$ 0.0293	2.2034	4.7593	18	–
	RAPT	0.3818 $\pm$ 0.0184	0.6172 $\pm$ 0.0127	0.4703 $\pm$ 0.0185	0.3582 $\pm$ 0.0192	0.2594	2.8257	2	16
	RAPT-Enhanced	0.3783 $\pm$ 0.0321	0.5677 $\pm$ 0.0222	0.4600 $\pm$ 0.0307	0.3614 $\pm$ 0.0318	2.6492	5.8812	2	16
5G NR Lat.	Frozen	0.8961 $\pm$ 0.0099	0.9075 $\pm$ 0.0061	0.9314 $\pm$ 0.0083	0.8798 $\pm$ 0.0085	0.0000	1.7862	0	–
	Event-Driven	0.8903 $\pm$ 0.0107	0.9055 $\pm$ 0.0060	0.9475 $\pm$ 0.0085	0.8688 $\pm$ 0.0083	1.0406	2.7561	11	–
	Full Retraining	0.9027 $\pm$ 0.0059	0.9145 $\pm$ 0.0045	0.9478 $\pm$ 0.0053	0.8829 $\pm$ 0.0055	0.7636	2.4689	9	–
	RAPT	0.8894 $\pm$ 0.0110	0.9025 $\pm$ 0.0068	0.9258 $\pm$ 0.0103	0.8737 $\pm$ 0.0092	0.3955	2.1112	3	6
	RAPT-Enhanced	0.8915 $\pm$ 0.0117	0.9055 $\pm$ 0.0076	0.9348 $\pm$ 0.0106	0.8737 $\pm$ 0.0098	0.8609	2.5710	3	6

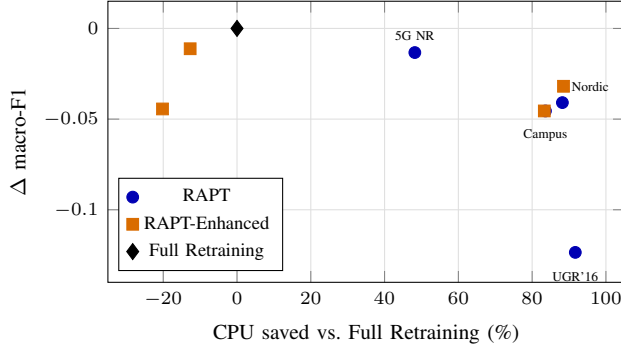


Fig. 4. Accuracy–cost trade-off on all four streams, measured against Full Retraining at the origin. Circles are RAPT and squares are RAPT-Enhanced, computed from Table II. Up and to the right is better. Squares left of zero cost more adaptation CPU than Full Retraining.

G. Efficiency Ablation

Table III and Fig. 5 show the mechanism ladder on 5G Campus, in which each row adds one mechanism. RAPT-Full is the base controller with the similarity gate, and it scores 0.9587 ± 0.0000 macro-F1 at 0.4283 s. Replacing the absolute refit trigger with a relative-degradation trigger (RAPT-Evidence) leaves macro-F1 unchanged and fires *zero* refreshes. With an absolute floor (RAPT-Floor), 7.6 refreshes fire and the score is 0.9797 ± 0.0050 at 0.5865 s, higher than before but noisier. RAPT-Cheap keeps the periodic schedule and uses the cheaper refit. Its score is 0.9851 ± 0.0029 at 0.8477 s, compared with 0.9829 ± 0.0023 at 1.4063 s for Full Retraining. The macro-F1 difference is $\Delta = +0.0022$ with $p = 0.3125$, $d = 0.47$, and because the 95% interval $[-0.0036, +0.0080]$ includes zero we cannot claim a difference. The periodic refresh with a full refit scores 0.9847 ± 0.0028 at 2.7168 s. Set against it, the cheap refit keeps the accuracy and cuts

TABLE III

PERFORMANCE OF DIFFERENT CONFIGURATIONS OF RAPT USING THE 5G CAMPUS STREAM, AVERAGED OVER MULTIPLE RANDOM SEEDS.

Configuration	Macro-F1	Adapt. CPU	Refreshes
Full Retraining (ref.)	0.9829 $\pm$ 0.0023	1.4063	–
RAPT-Full	0.9587 $\pm$ 0.0000	0.4283	0
RAPT-Evidence	0.9587 $\pm$ 0.0000	0.4260	0
RAPT-Floor	0.9797 $\pm$ 0.0050	0.5865	7.6
RAPT-Cheap	0.9851 $\pm$ 0.0029	0.8477	22
RAPT-Incremental	0.9549 $\pm$ 0.0020	0.5813	0

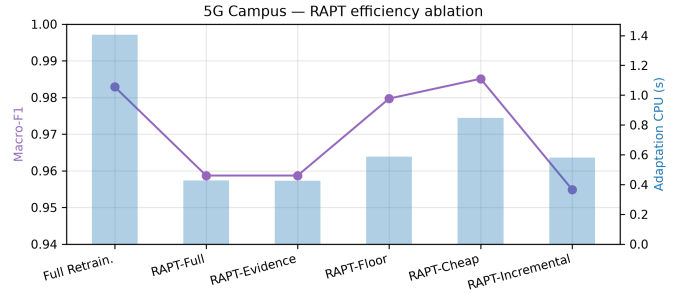


Fig. 5. Ablation ladder comparing the different configurations of RAPT, evaluated on the 5G Campus stream. As shown, RAPT-Cheap matches the full refitter in terms of macro-F1 at around 40% less adaptation CPU, whereas RAPT-Incremental performs worse on both accuracy and cost.

adaptation CPU by roughly 69%. RAPT-Incremental scores 0.9549 ± 0.0020 at 0.5813 s. Against RAPT-Full (0.9587 at 0.4283 s) it is worse on both axes: lower macro-F1 and higher adaptation CPU. It is therefore not carried forward. The mechanism that does survive is the periodic schedule with the cheaper refit, RAPT-Cheap, which is the only configuration here that combines accuracy on par with Full Retraining and a clear cost reduction.

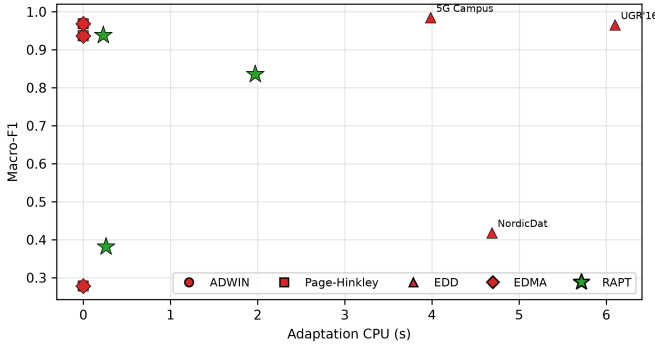


Fig. 6. Historical detectors. ADWIN, Page-Hinkley and EDMA never fire, so they replicate the frozen baseline. EDD fires most often, reaching the highest macro-F1 at the highest adaptation cost. RAPT adapts less often than all but two methods, and less accurately.

H. Determinants of Cost and Trigger Behavior

The cheap refresh follows direct refit-timing measurements with the ensemble held fixed: a 20-tree refit takes about 22 ms against 104 ms for 100 trees, and a 200-observation buffer about 80 ms against 104 ms for 1000. Tree count dominates, so the cheap refit cuts refit time by about four-fifths.

The zero-refresh result for RAPT-Evidence is a property of the mechanism, not a tuning outcome. A relative trigger asks whether current accuracy has fallen by more than a margin against a decayed reference built from that policy’s own earlier accuracy. When a regime is persistently poor rather than deteriorating, the reference decays toward the poor level, the gap the rule tests for never opens, and the trigger stays silent even though the policy is unfit. RAPT-Floor tests the same condition against a fixed floor and fires 7.6 refreshes where the relative rule fires none; that floor is stream-specific and would need calibration elsewhere.

I. Comparison to Historical Detectors

Of the four detectors, three never fire: ADWIN, Page-Hinkley and EDMA recorded zero adaptation events on all three streams and so reproduce the frozen baseline by construction. EDD fired 38–39 times per stream and reached the highest macro-F1 of any method on 5G Campus (0.9847 ± 0.0019) and NordicDat (0.4184 ± 0.0165), at the highest adaptation cost (3.98, 6.10 and 4.69 s). RAPT fires 2–11 times per stream depending on configuration, at lower adaptation cost than EDD on all three datasets (0.2292 vs. 3.98 s on Campus, 1.9711 vs. 6.10 s on UGR’16, 0.2594 vs. 4.69 s on NordicDat), and on Campus within 0.002 macro-F1 of EDD. On UGR’16 and NordicDat the cheaper trigger also yields lower macro-F1, so RAPT is the lowest-cost adaptive option here rather than the most accurate one.

VI. DISCUSSION

The cost saving appears wherever reuse occurs, but the accuracy penalty is concentrated on UGR’16. There a reused policy fits poorly, and enlarging the refit buffer takes 0.8360 to 0.9276; the parity check the design relied on never fires. Reuse

should therefore come with a check on the reused policy’s recent performance, but our runs do not show that the parity check we implemented is that check. Reusing on the regime identifier alone is unsafe when a returning regime no longer behaves as it did when its policy was stored; our evidence on UGR’16 does not separate a moved mapping from moved features.

VII. INDEPENDENT RE-VALIDATION

Every number here was re-derived from raw per-window records by a harness that re-implements the streaming protocol rather than reading the committed aggregates. It reproduces Tables II and III to four decimal places on every predictive metric, and an automated check of 205 quantitative claims against the committed artifacts passes in full. Three findings are worth recording.

Cost is machine-dependent. A re-run of the 9B drift pipeline reproduced every F1, accuracy, precision and recall exactly but shifted adaptation CPU by up to 0.14 s; the ordering is stable, the absolute seconds are not.

Retrieval keys do not track behaviour. An oracle retraining on the previous visit of the *same* regime reaches only 0.5926 ± 0.0077 macro-F1 on UGR’16 against 0.9595 for Full Retraining, but 0.9905 ± 0.0014 on Campus: regime identity survives on Campus and not on UGR’16, exactly where blind reuse collapses. Whether the UGR’16 failure is a moved label mapping or a covariate shift is not settled here, since the conditional-vs-covariate test is inconclusive on that stream and a frozen model’s AUC is a weak probe of $P(Y | X)$.

Controls. Equivalence to Full Retraining on Campus holds at a ± 0.010 macro-F1 margin ($p = 0.0102$) but not at ± 0.005 ($p = 0.13$): bounded equivalence within about one F1 point, not “within measurement noise”. A label-delay variant, where the target arrives one window late, leaves Frozen and Event-Driven unchanged but costs RAPT 0.13 macro-F1 on NordicDat and Full Retraining 0.06.

VIII. LIMITATIONS AND THREATS TO VALIDITY

With five seeds the smallest attainable two-sided p -value in a seed-level comparison is 0.0625, so some large effects cannot reach formal significance; this applies most to the ablation. Window-level tests assume windows from one stream are independent, which inflates the effective sample size. UGR’16 per-window macro-F1 is metric-sensitive, and 7.6% of its stored policies carry a provenance mismatch we have not isolated. The mechanism relies on regime recurrence, which we measure and do not assume: without recurrence the repository is never exercised. The RAPT-Floor threshold and the 20-tree cheap refit were tuned on the evaluated stream. Finally, the repository is never pruned, and the approach treats regime similarity as meaningful.

IX. CONCLUSION

Retraining is the standard response to drift, but when conditions return, part of what was paid for is already in a previously trained model. RAPT keeps one policy per regime

and retrieves it on recurrence, training anew only when no suitable policy exists and refreshing a stored policy once monitoring shows it no longer fits. Across four network streams adaptation CPU was 48–92% below full retraining, while macro-F1 was lower on all four. Where a returning regime no longer matched its stored policy, blind reuse caused substantial damage; enlarging the refit buffer recovered most of it, while the parity check we implemented never fired. Policy reuse is worth deploying when regimes recur with consistent or monitored behaviour, and the next step is to test whether a retrieved regime still means what it did when its policy was stored.

REFERENCES

- [1] Ahmad Abbasi, Abdul Rehman Javed, Chinmay Chakraborty, Jamel Nebhen, Wisha Zehra, and Zunera Jalil. Elstream: An ensemble learning approach for concept drift detection in dynamic social big data stream learning. *IEEE Access*, 9:66408–66419, 2021.
- [2] Manuel Baena-García, José del Campo-Ávila, Raul Fidalgo, Albert Bifet, Ricard Gavaldà, and Rafael Morales-Bueno. Early drift detection method. In *Proc. Int. Workshop on Knowledge Discovery from Data Streams*, pages 77–86, 2006.
- [3] Roberto S. M. Barros, Danilo R. L. Cabral, Paulo M. Gonçalves, and Silas G. T. C. Santos. Rddm: Reactive drift detection method. *Expert Systems with Applications*, 90:344–355, 2017.
- [4] Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proc. SIAM Int. Conf. Data Mining (SDM)*, pages 443–448, 2007.
- [5] Tobias Brockhoff, Merih Seran Uysal, and Wil M. P. van der Aalst. Time-aware concept drift detection using the earth mover’s distance. In *Proc. Int. Conf. Process Mining (ICPM)*, pages 33–40, 2020.
- [6] Dariusz Brzeziński and Jerzy Stefanowski. Reacting to different types of concept drift: The accuracy updated ensemble algorithm. *IEEE Transactions on Neural Networks and Learning Systems*, 25(1):81–94, 2014.
- [7] Pedro Domingos and Geoff Hulten. Mining high-speed data streams. In *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pages 71–80, 2000.
- [8] Denis Moreira dos Reis, Peter Flach, Stan Matwin, and Gustavo Batista. Fast unsupervised online drift detection using incremental kolmogorov–smirnov test. In *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pages 1545–1554, 2016.
- [9] Isvani Frias-Blanco, José del Campo-Ávila, Gonzalo Ramos-Jimenez, Rafael Morales-Bueno, Agustín Ortiz-Díaz, and Yailé Caballero-Mota. Online and non-parametric drift detection methods based on hoeffding’s bounds. *IEEE Transactions on Knowledge and Data Engineering*, 27(3):810–823, 2015.
- [10] Joao Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Proc. Brazilian Symp. Artificial Intelligence (SBIA)*, pages 286–295, 2004.
- [11] Heitor M. Gomes, Albert Bifet, Jesse Read, Jean Paul Barddal, Fabrício Enembreck, Bernhard Pfahringer, Geoff Holmes, and Talel Abdesslem. Adaptive random forests for evolving data stream classification. *Machine Learning*, 106(9–10):1469–1495, 2017.
- [12] David P. Helmbold and Philip M. Long. Tracking drifting concepts by minimizing disagreements. *Machine Learning*, 14(1):27–45, 1994.
- [13] Imen Khamassi, Moamar Sayed-Mouchaweh, Moez Hammami, and Khaled Ghédira. Discussion and review on evolving data streams and concept drift adapting. *Evolving Systems*, 9(1):1–23, 2018.
- [14] J. Zico Kolter and Marcus A. Maloof. Dynamic weighted majority: An ensemble method for drifting concepts. *Journal of Machine Learning Research*, 8(91):2755–2790, 2007.
- [15] Jie Lu, Anjin Liu, Fan Dong, Feng Gu, Joao Gama, and Guangquan Zhang. Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, 31(12):2346–2363, 2018.
- [16] Shaul Markovitch and Paul D. Scott. The role of forgetting in learning. In *Machine Learning Proceedings 1988*, pages 459–465. Morgan Kaufmann, 1988.
- [17] Topi Miekka, Pasi Pyykönen, Georgios Drainakis, Panagiotis Pantazopoulos, Tobias Müller, Konstantinos V. Katsaros, Vasilis Sourlas, Angelos Amditis, and Dimitra I. Kaklamani. Nordicdat: A cross-border predictive qos dataset. In *Proc. IEEE Global Communications Conf. (GLOBECOM)*, pages 1281–1286, 2024. Dataset: Zenodo 10.5281/zenodo.10964584, CC-BY-4.0.
- [18] Jacob Montiel, Max Halford, Saulo Martiello Mastelini, Geoffrey Bolmier, Raphael Sourty, Robin Vaysse, Adil Zouitine, Heitor Murilo Gomes, Jesse Read, Talel Abdesslem, and Albert Bifet. River: machine learning for streaming data in python. *Journal of Machine Learning Research*, 22(110):1–8, 2021.
- [19] Ewan S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.
- [20] Marcin Pakuła and Paweł Kryszkiewicz. 5g nr end-to-end latency simulation dataset. Zenodo record 20035549, CC-BY-4.0, 2026.
- [21] Sinno Jialin Pan and Qiang Yang. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2010.
- [22] Gordon J. Ross, Niall M. Adams, Dimitris K. Tasoulis, and David J. Hand. Exponentially weighted moving average charts for detecting concept drift. *Pattern Recognition Letters*, 33(2):191–198, 2012.
- [23] Jeffrey C. Schlimmer and Richard H. Granger. Incremental learning from noisy data. *Machine Learning*, 1(3):317–354, 1986.
- [24] Milind Tambe and Allen Newell. Some chunks are expensive. In *Machine Learning Proceedings 1988*, pages 451–458. Morgan Kaufmann, 1988.
- [25] Gerhard Widmer. Combining robustness and flexibility in learning drifting concepts. In *Proc. European Conf. Artificial Intelligence (ECAI)*, pages 468–472, 1994.
- [26] Li Yang, Dimitrios Michael Manias, and Abdallah Shami. Pwpae: An ensemble framework for concept drift adaptation in iot data streams. In *Proc. IEEE Global Communications Conf. (GLOBECOM)*, pages 1–6, 2021.
- [27] Li Yang and Abdallah Shami. A lightweight concept drift detection and adaptation framework for iot data streams. *IEEE Internet of Things Magazine*, 4(2):96–101, 2021.